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Course : Artificial Intelligence
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Date : 27/07/2026

1. Smart Home Automation System - Types of Intelligent Agent :

Introduction :

A smart home automation system controls lighting, temperature and security based on user behaviour and environmental conditions, it receives inputs from sensors such as motion detectors, temperature sensors, and clocks and it performs actions automatically to improve comfort, safety and energy efficiency.

Types :-

1. Simple Reflex Agent :

Definition :

A simple reflex agent works according to predefined condition-action rules. It only considers the current input.

Working in Smart Home :

- * Motion detected - Turn off lights.
- * No motion for 10 minutes - Turn OFF lights.
- * Temperature $> 30^{\circ}\text{C}$ - Turn ON fan or AC.

Advantages:-

- * Fast response.
- * Easy to implement.
- * Low computational cost.

Disadvantages:-

- * Does not remember previous event.
- * Cannot learn user preferences.
- * Not suitable for complex situations.

2. Model-Based Reflex Agent:-

Definition:

A model-based agent maintains an internal model of the environment and uses past information along with current inputs.

Working:

- * Knows whether someone is inside the home.
- * keeps track of room occupancy.
- * Detects abnormal situations in the home.

Advantages:

- * Handles partially.
- * Better decision making.
- * More reliable than simple reflex Agent.

Disadvantages:-

- * Requires more Memory.
- * More complex implementation.

Conclusion:

An hybrid intelligent agent provides the best performances for smart homes by combining fast response, planning, memory and optimization. It ensures maximum comfort, energy efficiency and security.

2. Robot is an Unknown maze - UCS and TDS:

Introduction:

A robot placed in an unknown maze and must search for a path to exit without prior knowledge of the layout.

1. Uniform Cost Search (UCS):

UCS expands the node with the lowest cumulative path cost first.

Working:

- * Start from the initial position.
- * Calculate the total cost of each path.
- * Expand the path with the smallest cost.
- * Continue until the exit is found.

Advantages:

- * Complete.
- * Finds the least-cost path.

Disadvantages:

- * Uses large memory.
- * Slower for large search spaces.

Complexity:

* Time: $O(b^{(1+LC/\epsilon)})$

* Space: $O(b^{(1+LC/\epsilon)})$

where,

b = branching factor

C^* = optimal path cost

ϵ = minimum step cost

3. Goal-Based Agent:

Definition:

A Goal-Based Agents chooses actions that helps achieve a specific goal.

Working:

- * Maintain room temperature at 24°C .

- * Keep home secure.

- * It has a goal to protect.

Advantages:

- * It keeps the home secure.

- * More reliable than Model-Based.

Disadvantages:

- * Not suitable for complex.

- * Implementation takes more time.

4. Utility-based agents:

Definition:

The utility-Based agents find the shortest path and secure exit in home.

Working:

- * It also keeps the home secure.

- * Detects abnormal situations.

Advantages:

- * Better decision making.

Disadvantages:

- * More complex time.

2. Iterative Deepening Search (IDS):

Definition:

IDS repeatedly performs Depth-Limited search with increasing depth limits until the goal is found.

Working:

- * Search depth = 0.
- * If not found, increase depth to 1.
- * Continue increasing depth until exit.

Advantages:

- * Low memory usage.
- * Complete.
- * Finds shortest path.

Disadvantages:

- * Repeats node expansion.
- * Not suitable.

Complexity:

- * Time: $O(b^d)$
- * Space: $O(bd)$

Best Combination:

Goal-Based Agent + Uniform Cost Search (UCS)

Justification:-

- * Goal-Based focuses on reaching the exit.
- * UCS guarantees the lowest-cost path.
- * Suitable when movement costs differ.

Goal-Based Agent + IDS is also a good choice.

Conclusion:

For an unknown maze, a Goal-Based Agent using uniform Cost search is the most effective solution because it finds the optimal path while making intelligent decisions.